

Optimality conditions: linear optimization

Strong duality and complementarity slackness

Michel Bierlaire

Optimization: principles and algorithms



Strong duality and complementarity slackness

We saw that an optimal solution of the primal is obtained when a basis is available such that all reduced costs are non negative

We show here that the same basis can be used to obtain an optimal solution of the dual problem.

Context

Data

$$A \in \mathbb{R}^{m \times n}$$

$$b \in \mathbb{R}^m$$

$$c \in \mathbb{R}^n$$

Context

Data

$$A \in \mathbb{R}^{m \times n}$$

$$b \in \mathbb{R}^m$$

$$c \in \mathbb{R}^n$$

Primal

$$\min_{x \in \mathbb{R}^n} c^T x$$

subject to

$$Ax = b$$

$$x \geq 0$$

Context

Data

$$A \in \mathbb{R}^{m \times n}$$

$$b \in \mathbb{R}^m$$

$$c \in \mathbb{R}^n$$

Primal

$$\min_{x \in \mathbb{R}^n} c^T x$$

subject to

$$Ax = b$$

$$x \geq 0$$

Dual

$$\max_{\lambda \in \mathbb{R}^m} \lambda^T b$$

subject to

$$A^T \lambda \leq c$$

Optimal solutions: primal

Assumptions

$$A = (B|N)$$

$$B^{-1}b \geq 0$$

$$\bar{c} = c - A^T B^{-T} c_B \geq 0$$

Optimal solutions: primal

Assumptions

$$A = (B|N)$$

$$B^{-1}b \geq 0$$

$$\bar{c} = c - A^T B^{-T} c_B \geq 0$$

Primal solution

$$x_B^* = B^{-1}b, \quad x_N^* = 0.$$

Optimal solutions: dual

Assumptions

$$A = (B|N)$$

$$B^{-1}b \geq 0$$

$$\bar{c} = c - A^T B^{-T} c_B \geq 0$$

Optimal solutions: dual

Assumptions

$$A = (B|N)$$

$$B^{-1}b \geq 0$$

$$\bar{c} = c - A^T B^{-T} c_B \geq 0$$

Dual solution

$$\lambda^* = B^{-T} c_B.$$

Optimal solutions: dual

Assumptions

$$A = (B|N)$$

$$B^{-1}b \geq 0$$

$$\bar{c} = c - A^T B^{-T} c_B \geq 0$$

λ^* is dual-feasible

$$A^T \lambda^* = A^T B^{-T} c_B = c - \bar{c} \leq c$$

Dual solution

$$\lambda^* = B^{-T} c_B.$$

Optimal solutions: dual

Assumptions

$$A = (B|N)$$

$$B^{-1}b \geq 0$$

$$\bar{c} = c - A^T B^{-T} c_B \geq 0$$

λ^* is dual-feasible

$$A^T \lambda^* = A^T B^{-T} c_B = c - \bar{c} \leq c$$

Dual solution

$$\lambda^* = B^{-T} c_B.$$

λ^* is dual-optimal

$$(\lambda^*)^T b = c_B^T B^{-1} b = c_B^T x_B^* = c^T x^*$$

Strong duality

- ▶ If the primal or the dual has an optimal solution,
- ▶ so does the other one,
- ▶ and the optimal objective values are equal.

Complementarity slackness

$$c^T x^* = (\lambda^*)^T b$$

$$c^T x^* = (\lambda^*)^T b$$

$$c^T x^* = (\lambda^*)^T A x^*$$

$$c^T x^* - (\lambda^*)^T A x^* = 0$$

$$(c - A^T \lambda^*)^T x^* = 0$$

Complementarity slackness

$$(c - A^T \lambda^*)^T x^* = \sum_{i=1}^n (c_i - \sum_{j=1}^m a_{ji} \lambda_j^*) x_i^* = 0$$

- ▶ As all terms involved are non negative, each term must be 0.

Complementarity slackness

$$(c - A^T \lambda^*)^T x^* = \sum_{i=1}^n (c_i - \sum_{j=1}^m a_{ji} \lambda_j^*) x_i^* = 0$$

- ▶ As all terms involved are non negative, each term must be 0.
- ▶ Consider an index $i = 1, \dots, n$.

Complementarity slackness

$$(c - A^T \lambda^*)^T x^* = \sum_{i=1}^n (c_i - \sum_{j=1}^m a_{ji} \lambda_j^*) x_i^* = 0$$

- ▶ As all terms involved are non negative, each term must be 0.
- ▶ Consider an index $i = 1, \dots, n$.
- ▶ Consider the corresponding primal constraint

$$x_i \geq 0.$$

Complementarity slackness

$$(c - A^T \lambda^*)^T x^* = \sum_{i=1}^n (c_i - \sum_{j=1}^m a_{ji} \lambda_j^*) x_i^* = 0$$

- ▶ As all terms involved are non negative, each term must be 0.
- ▶ Consider an index $i = 1, \dots, n$.
- ▶ Consider the corresponding primal constraint

$$x_i \geq 0.$$

- ▶ Consider the corresponding dual constraint

$$\sum_{j=1}^m a_{ji} \lambda_j \leq c_i.$$

Complementarity slackness

$$(c - A^T \lambda^*)^T x^* = \sum_{i=1}^n (c_i - \sum_{j=1}^m a_{ji} \lambda_j^*) x_i^* = 0$$

- ▶ As all terms involved are non negative, each term must be 0.
- ▶ Consider an index $i = 1, \dots, n$.
- ▶ Consider the corresponding primal constraint

$$x_i \geq 0.$$

- ▶ Consider the corresponding dual constraint

$$\sum_{j=1}^m a_{ji} \lambda_j \leq c_i.$$

- ▶ At least one of them must be active at the solution.

Summary

A feasible basic solution such that the reduced costs are all non negative provides the optimal solution for both the primal and the dual problems

The strong duality theorem states than, if one of the two problems has an optimal solution, so does the other one, and the objective functions coincide.

The complementarity slackness states that if a constraint is not active in one problem, the corresponding variable in the other problem must be zero.